Day 9: Subnetting Speed Drills

Level 1, Month 1, Networking foundations

What I learned

- The mistakes today were about rushing, not about understanding. The method works when I slow down.
- Goal for now: 3 problems in a row with no mistakes. Speed comes later from correct repetition.
- Bits and block size are different things:

```
bits = the exponent
taille du bloc = 2 ^ bits
hotes valides = 2 ^ bits - 2
```

- Three stops for every problem:
  1. Write the mask in big writing at the top of the page.
  2. Check every power by multiplying (2^3 = 2 x 2 x 2 = 8).
  3. Before sending: the given address is inside the block, derniere is smaller than broadcast, and hosts + 2 = 2^bits.
- Subtraction trick: count upward from the smaller number and check by adding. For example 57 to 64: 3 + 4 = 7, and 57 + 7 = 64.

Problems solved

Address	Taille	Reseau	Premiere	Broadcast	Dernie
172.18.201.30/20	16	172.18.192.0	172.18.192.1	172.18.207.255	172.18
192.168.50.100/27	32	192.168.50.96	192.168.50.97	192.168.50.127	192.16
10.4.77.9/21	8	10.4.72.0	10.4.72.1	10.4.79.255	10.4.7
172.16.130.200/25	128	172.16.130.128	172.16.130.129	172.16.130.255	172.16

Review question (from Day 7): `TTL=57` means probably not Windows (start 64), and 64 - 57 = 7 routers.

Error log

Mistake	Correct answer	Why
172.18.201.30/20: solved it as /30	/20	I misread the mask. Fix: write the mask in big writing at the top.
Hosts 2096	4094	Subtraction error. 2^12 = 4096, minus 2. Check: 4094 + 2 = 4096.
TTL 57: 13 routers	7 routers	I subtracted column by column without borrowing. Count upward and check by adding.
192.168.50.100/27: used 5 as the step	32	5 is the number of bits. The block size is 2^5 = 32, and the block starts at 96.
10.4.77.9/21: wrote 2^3 = 16	2^3 = 8	16 is 2^4. The blocks go by 8, so 77 is in the block 72 to 79.

Main lessons

- bits = the exponent, taille = 2^bits. I mixed them twice today.
- Writing the mask at the top of the page fixes the misreading.
- A clear page with one step per line prevents reading errors, especially when I cross things out.
- Time: 3 problems in 6 minutes on the first try, but with mistakes. Accuracy first.

Next session

Introduction to VLSM: cutting one network into subnets of different sizes.